Litong (Tony) Xing

New York, USA | tong_xing666@126.com

EDUCATION

The Stony Brook School | New York, USA

August 2023-present | High school diploma expected May 2027

Coursework includes AP Calculus BC, college multivariable calculus, and advanced linear algebra. SAT: 1520 (Reading and Writing 730; Math 790).

SELECTED HONORS

USAMO qualifier; USACO Platinum Division; AMC 12A perfect score (150/150) (Grade 11). HiMCM 2025 Honorable Mention. PBIC Team First Prize and China Conrad Challenge finalist (Grade 10). AIME qualification in Grades 9-11.

RESEARCH

Differential Privacy

Grades 11-12 | Mentor: Professor Andrew Blumberg, Columbia University

Compared local and central differential privacy using Python simulations of 10,000 users, 88,000+ ad impressions, and eight campaigns. Evaluated three mechanisms across privacy budgets, dataset sizes, and click rates, including accuracy, campaign ranking, and attribute-inference risks. Wrote an 8,000+ word paper published in NHSJS.

Published article: Local and Central Differential Privacy in Online Advertising

Distributed Transactions & Geo-Replication

Summer 2026 | CSIRE, Stony Brook University | Professor Shuai Mu

Studied Mako: Speculative Distributed Transactions with Geo-Replication. Modeled and compared transaction systems, traditional two-phase commit, geo-replication, and speculative execution. Presented a final paper, poster, and symposium presentation.

Graph Decomposition & Edge Contraction

Summer 2025 | Pioneer Academics

Conducted independent research on biclique decompositions and edge contraction. Investigated how edge contraction affects the minimum number of complete bipartite graphs needed to decompose a graph; wrote a research paper.

Environmental Footprint of Major Sporting Events

Grade 11 | HiMCM | Two-person team leader

Modeled energy use, greenhouse gas emissions, water, and waste. Applied hierarchical sub-models and an entropy/TOPSIS framework to rank host cities. Used Super Bowl LIX data and extended the framework to the BWF World Tour. Wrote a 25-page report.

Research, experience & community

Litong (Tony) Xing

ADDITIONAL PROJECTS

High-Performance Computing Environmental Footprint

Grade 10 | Two-person team leader

Studied global data centers' energy consumption, carbon emissions, and water use. Developed ARIMA and LSTM forecasting models and proposed sustainable solutions. Wrote a 20+ page report.

Medicine AI | Conrad Challenge

Grade 10 | Student developer

Developed a medication-assistance prototype for older adults and chronic-disease patients. Integrated OCR and TextCNN, designed reminders and caregiver reports, and built a PyQt5 interface; tested the system using medical data.

EXPERIENCE & SUMMER LEARNING

Rongdingyue (Zhejiang) Aviation Technology Co., Ltd.

Summer 2026 | UAV Technology Data Analysis & International Business Intern

Completed a two-week internship: revised the English product manual, operated UAVs, managed flight-test data, evaluated payload capacity, and conducted pre-delivery test flights.

AwesomeMath (Summer 2024): Computational Geometry - Level 2. Ivy League Summer School (Summer 2023): English literature, writing, and public speaking.

LEADERSHIP & COMMUNITY

Asian Student Union | President

Grade 12; Executive Board, Grade 11; Member, Grades 9-10

Lead a 10-member executive board and 30+ members. Organized six cultural events that generated $2,000+ for ASU programs.

Chinese Leader & Chinese Teaching Program

Chinese Leader, Grade 12 | Teaching Assistant, Grade 9-present

Support 100+ Chinese students and help revise the school's Chinese-language handbook. Assist twice-monthly Chinese language and culture classes for approximately 20 elementary students.

Sports-Related Injuries Program | Co-leader

Grades 9-11

Raised approximately $5,000 for protective equipment for youth athletes. Partnered with two soccer clubs in Arusha, Tanzania, and local medical professionals on equipment distribution and injury-prevention resources.

Other campus roles: E-sports Club vice president; Chess Club member (school team county champions in Grades 9 and 10); admissions tour guide (10+ tours); Kitchen Crew Captain (leading 20 students).

SKILLS & INTERESTS

Technical: C++, Python, mathematical modeling, data analysis.
Languages: Mandarin (native), English (fluent), French (intermediate).
Interests: Badminton, basketball, cycling, photography, piano (Level 7), taekwondo (blue-red belt).